

3. Pricing of Bonds

1. A pension fund manager invests \$10 million in a debt obligation that promises to pay 7.3% per year for four years. What is the future value of the \$10 million?

To determine the future value of any sum of money invested today, we can use the future value equation, which is: $P_n = P_0 (1 + r)^n$ where n = number of periods, P_n = future value n periods from now (in dollars), P_0 = original principal (in dollars) and r = interest rate per period (in decimal form). Inserting in our values, we have: $P_4 = \$10,000,000(1.073)^4 = \$10,000,000(1.325558466) = \$13,255,584.66$.

2. Suppose that a life insurance company has guaranteed a payment of \$14 million to a pension fund 4.5 years from now. If the life insurance company receives a premium of \$10.4 million from the pension fund and can invest the entire premium for 4.5 years at an annual interest rate of 6.25%, will it have sufficient funds from this investment to meet the \$14 million obligation?

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3. Answer the below questions.

(a) The portfolio manager of a tax-exempt fund is considering investing \$500,000 in a debt instrument that pays an annual interest rate of 5.7% for four years. At the end of four years, the portfolio manager plans to reinvest the proceeds for three more years and expects that for the three-year period, an annual interest rate of 7.2% can be earned. What is the future value of this investment?

At the end of year four, the portfolio manager's amount is given by: $P_n = P_0 (1 + r)^n$. Inserting in our values, we have $P_4 = \$500,000(1.057)^4 = \$500,000(1.248245382) = \$624,122.66$. In three more years at the end of year seven, the manager amount is given by: $P_7 = P_4(1 + r)^3$. Inserting in our values, we have: $P_7 = \$624,122.66(1.072)^3 = \$624,122.66(1.231925248) = \$768,872.47$.

(b) Suppose that the portfolio manager in Question 3, part a, has the opportunity to invest the \$500,000 for seven years in a debt obligation that promises to pay an annual interest rate of 6.1% compounded semiannually. Is this investment alternative more attractive than the one in Question 3, part a?

At the end of year seven, the portfolio manager's amount is given by the following equation, which adjusts for semiannual compounding. We have: $P_n = P_0(1 + r/2)^{2(n)}$. Inserting in our values, we have $P_7 = \$500,000(1 + 0.061/2)^{2(7)} = \$500,000(1.0305)^{14} = \$500,000(1.522901960) = \$761,450.98$. Thus, this investment alternative is not more attractive. It is less by the amount of $\$761,450.98 - \$768,872.47 = -\$7,421.49$.

4. Suppose that a portfolio manager purchases \$10 million of par value of an eight-year bond that has a coupon rate of 7% and pays interest once per year. The first annual coupon payment will be made one year from now. How much will the portfolio manager have if she (1) holds the bond until it matures eight years from now, and (2) can reinvest all the annual interest payments at an annual interest rate of 6.2%?

At the end of year eight, the portfolio manager's amount is given by the following equation, which adjusts for annual compounding.

We have:

$$P_n = A \left[\frac{(1+r)^n - 1}{r} \right] + \text{Par Value}$$

where A = coupon rate times par value. Inserting in our values, we have:

$$P_8 = 0.07(\$10,000,000) \left[\frac{(1+0.062)^8 - 1}{0.062} \right] + \$10,000,000 = \$700,000[9.9688005] + \$10,000,000 = \$6,978,160.38 + \$10,000,000 = \$16,978,160.38.$$

5. Answer the below questions.

(a) If the discount rate that is used to calculate the present value of a debt obligation's cash flow is increased, what happens to the price of that debt obligation?

A fundamental property of a bond is that its price changes in the opposite direction from the change in the required yield. The reason is that the price of the bond is the present value of the cash flows. As the required yield increases, the present value of the cash flow decreases; hence the price decreases. The opposite is true when the required yield decreases: The present value of the cash flows increases, and therefore the price of the bond increases.

(b) Suppose that the discount rate used to calculate the present value of a debt obligation's cash flow is $x\%$. Suppose also that the only cash flows for this debt obligation are \$200,000 four years from now and \$200,000 five years from now. For which of these cash flows will the present value be greater?

Cash flows that come earlier will have a greater value. As long as $x\%$ is positive and the amount is the same, the present value will be greater for the \$200,000 four years from now compared to five years from now. This can also be seen by noting that if $x >$

0 then $\left[\frac{1}{(1+x)^4} \right] > \left[\frac{1}{(1+x)^5} \right]$. The latter inequality implies

$\$2,000 \left[\frac{1}{(1+x)^4} \right] > \$2,000 \left[\frac{1}{(1+x)^5} \right]$ will hold.

6. Calculate for each of the following bonds the price per \$1,000 of par value assuming semiannual coupon payments.

Bond	Coupon Rate (%)	Years to Maturity	Required Yield (%)
A	9	20	9
B	6	15	10
C	0	14	8

Consider a 20-year 9% coupon bond with a par value of \$1,000 and a required yield of 9%. Given $C = 0.09(\$1,000) / 2 = \45 , $n = 2(20) = 40$ and $r = 0.09 / 2 = 0.045$, the present value of the coupon payments is:

$$P = C \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right] = \$45 \left[\frac{1 - \frac{1}{(1.045)^{40}}}{0.045} \right] = \$45 \left[\frac{1 - \frac{1}{5.81863645}}{0.045} \right] =$$

$$\$45 \left[\frac{1 - 0.1719287}{0.045} \right] =$$

$$\$45[18.401584] = \$828.071.$$

The present value of the par or maturity value of \$1,000 is: $\frac{M}{(1+r)^n} = \frac{\$1,000}{(1.045)^{40}} =$

$$\frac{\$1,000}{5.81863645}$$

$= \$171.929$. Thus, the price of the bond (P) = $\$828.071 + \$171.929 = \mathbf{\$1,000.00}$. [NOTE. We already knew the answer would be \$1,000 because the coupon rate equals the yield to maturity.]

Consider a 15-year 6% coupon bond with a par value of \$1,000 and a required yield of 10%. Given $C = 0.06(\$1,000) / 2 = \30 , $n = 2(15) = 30$ and $r = 0.10 / 2 = 0.05$, the present value of the coupon payments is:

$$P = C \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right] = \$30 \left[\frac{1 - \frac{1}{(1.05)^{30}}}{0.05} \right] = \$30 \left[\frac{1 - \frac{1}{4.3219424}}{0.05} \right] =$$

$$\$30 \left[\frac{1 - 0.2313774}{0.05} \right] =$$

$$\$30[15.372451] = \$461.174.$$

The present value of the par or maturity value of \$1,000 is: $\frac{M}{(1+r)^n} = \frac{\$1,000}{(1.05)^{30}} =$

$$\frac{\$1,000}{4.3219424} =$$

\$231.377. Thus, the price of the bond (P) = \$461.174 + \$231.377 = **\$692.55**.

Consider a 14-year 0% coupon bond with a par value of \$1,000 and a required yield of 8%. Given $C = 0(\$1,000) / 2 = \0 , $n = 2(14) = 28$ and $r = 0.08 / 2 = 0.04$, the present value of the coupon payments is:

$$P = C \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right] = \$0 \left[\frac{1 - \frac{1}{(1.04)^{28}}}{0.04} \right] = \$0 \left[\frac{1 - \frac{1}{2.998703319}}{0.055} \right] =$$

$$\$0 \left[\frac{1 - 0.33477471}{0.055} \right] =$$

$\$0[16.66306322] = \0 . [NOTE. We already knew the answer because the coupon rate is zero.]

The present value of the par or maturity value of \$1,000 is: $\frac{M}{(1+r)^n} = \frac{\$1,000}{(1.04)^{28}} =$

$$\frac{\$1,000}{2.99870332} = \$333.48. \text{ Thus, the price of the bond (P) } = \$0 + \$333.48 = \textbf{\$333.48}.$$

7. Consider a bond selling at par (\$100) with a coupon rate of 6% and 10 years to

maturity.

(a) What is the price of this bond if the required yield is 15%?

We have a 10-year 6% coupon bond with a par value of \$1,000 and a required yield of 15%. Given $C = 0.06(\$1,000) / 2 = \30 , $n = 2(10) = 20$ and $r = 0.15 / 2 = 0.075$, the present value of the coupon payments is:

$$P = C \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right] = \$30 \left[\frac{1 - \frac{1}{(1.075)^{20}}}{0.075} \right] = \$30 \left[\frac{1 - \frac{1}{4.2478511}}{0.075} \right] =$$
$$\$30 \left[\frac{1 - 0.2354131}{0.075} \right] =$$

$$\$30[10.1944913] = \$305.835.$$

The present value of the par or maturity value of \$1,000 is: $\frac{M}{(1+r)^n} = \frac{\$1,000}{(1.075)^{20}} =$

$$\frac{\$1,000}{4.2478511} =$$

$$\$235.413. \text{ Thus, the price of the bond (P) = } \$305.835 + \$235.413 = \textbf{\$541.25}.$$

(b) What is the price of this bond if the required yield increases from 15% to 16%, and by what percentage did the price of this bond change?

If the required yield increases from 15% to 16%, then we have:

$$P = C \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right] = \$30 \left[\frac{1 - \frac{1}{(1.08)^{20}}}{0.08} \right] = \$30 [9.8181474] = \$294.544.$$

The present value of the par or maturity value of \$1,000 is: $\frac{M}{(1+r)^n} = \frac{\$1,000}{(1.08)^{20}} =$

$$\$214.548.$$

$$\text{Thus, the price of the bond (P) = } \$294.544 + \$214.548 = \textbf{\$509.09}.$$

The bond price falls with percentage fall is equivalent to $\frac{\$509.09 - \$541.25}{\$541.25} = -0.059409$ or about **-5.94%**.

(c) What is the price of this bond if the required yield is 5%?

If the required yield is 5%, then we have:

$$P = C \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right] = \$30 \left[\frac{1 - \frac{1}{(1.025)^{20}}}{0.025} \right] = \$30 [15.5891623] = \$467.675.$$

The present value of the par or maturity value of \$1,000 is: $\frac{M}{(1+r)^n} = \frac{\$1,000}{(1.025)^{20}} = \$610.271$.

Thus, the price of the bond (P) = \$467.675 + \$610.271 = **\$1,077.95**.

(d) What is the price of this bond if the required yield increases from 5% to 6%, and by what percentage did the price of this bond change?

If the required yield increases from 5% to 6%, then we have:

$$P = C \left[\frac{1 - \frac{1}{(1+r)^n}}{r} \right] = \$30 \left[\frac{1 - \frac{1}{(1.03)^{20}}}{0.03} \right] = \$30 [14.87747486] = \$446.324.$$

The present value of the par or maturity value of \$1,000 is: $\frac{M}{(1+r)^n} = \frac{\$1,000}{(1.03)^{20}} = \$553.676$.

The price of the bond (P) = \$446.324 + \$553.676 = **\$1,000.00**. [NOTE. We already knew the answer would be \$1,000 because the coupon rate equals the yield to maturity.]

The bond price falls with the percentage fall equal to $(\$1,000.00 - \$1,077.95) / \$1,077.95 = -0.072310$ or **about -7.23%**.

(e) From your answers to Question 7, parts b and d, what can you say about the relative price volatility of a bond in a high-interest-rate environment compared to a low-interest-rate environment?

We can say that there is more volatility in a low-interest-rate environment because there was a greater fall (-7.23% versus -5.94%).

8. Suppose that you purchased a debt obligation three years ago at its par value of \$100,000 and nine years remaining to maturity. The market price of this debt obligation today is \$90,000. What are some reasons why the price of this debt obligation could have declined from time you purchased it three years ago?

The price of a bond will change for one or more of the following three reasons:

- (i) There is a change in the required yield owing to changes in the credit quality of the issuer.
- (ii) There is a change in the price of the bond selling at a premium or a discount, without any change in the required yield, simply because the bond is moving toward maturity.
- (iii) There is a change in the required yield owing to a change in the yield on comparable bonds (i.e., a change in the yield required by the market).

The first and third reasons are the likely reasons for the situation where the bond has plummeted from \$100,000 to \$90,000. The bond has plummeted in value because the credit quality of the issuer has fallen and/or the bond has plummeted because the yield on comparable bonds has increased.

9. Suppose that you are reviewing a price sheet for bonds and see the following prices (per \$100 par value) reported. You observe what seem to be several errors. Without calculating the price of each bond, indicate which bonds seem to be reported incorrectly, and explain why.

Bond	Price	Coupon Rate (%)	Required Yield (%)
U	90	6	9
V	96	9	8
W	110	8	6
X	105	0	5
Y	107	7	9
Z	100	6	6

If the required yield is the same as the coupon rate then the price of the bond should sell at its par value. This appears to be the case of bond Z since par values are typical at or near a \$100 quote. If the required yield decreases below the coupon rate then the price of a bond should increase. This is the case for bond W. This is not the case for bond V so this bond is not reported correctly. If the required yield increases above the coupon rate then the price of a bond should decrease. This is the case for bond U. This is not the case for bonds X and Y so these bonds are not reported correctly. Thus, bonds V, X, and Y are incorrectly reported because the change in the bond price is not consistent with the difference between the coupon rate and the required yield.

10. What is the “dirty” price of a bond?

The “dirty” (or “full”) price is the amount that the buyer agrees to pay the seller, which is the agreed-upon price plus accrued interest. The price of a bond without accrued interest is called the clean price. The exceptions are bonds that are in default. Such bonds are said to be quoted flat, that is, without accrued interest.